

Supplementary Material for “Statistical Certification of Autonomous Vehicle Fallback Triggers: Closed-Loop Evaluation Under Dataset Shift”

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This document contains the exploratory and post hoc analyses summarized in Section V of the main paper. All numbers use the same settings as the main paper ($\alpha = 0.05$, $\delta = 0.10$, 1,994 Argoverse 2 calibration and 1,996 test scenarios, 1,504 nuScenes scenarios) and can be regenerated from the released per-scenario records without re-simulation.

S1. CLASS-CONDITIONAL CERTIFICATION (POST HOC)

Table S1 implements the certificate of Luo *et al.* (cited in the main paper) in our framework: Mondrian split conformal prediction on the unsafe class, where unsafe means harm in the reference run and an alert means firing before it, with the finite-sample correction and without their tie randomization. At $\epsilon = 0.05$ on the 316 unsafe calibration scenarios it selects a threshold equivalent to always firing, so the class-conditional target is vacuous for this predictor at this level. The score separates harmful from harmless scenarios well enough for a marginal target (AUROC 0.83) but not for a 5 % miss rate among harmful ones. At the marginal-matched $\epsilon = \alpha/\pi = 0.32$ it selects the same threshold as empirical tuning. Deployed on nuScenes, that certificate has a conditional miss rate of 66 % against its 32 % target. The harm base rate rises from 15.8 % to 22.1 %, but the conditional miss rate is far above what that alone explains, so the class-conditional guarantee does not survive the dataset shift either.

TABLE S1

MARGINAL CERTIFICATE (LTT) VERSUS CLASS-CONDITIONAL CERTIFICATES (LUO-STYLE), THRESHOLDS CERTIFIED ON ARGOVERSE 2. MISS: MARGINAL MISS RATE; COND.: MISS RATE AMONG SCENARIOS WITH HARM IN THE REFERENCE RUN. RATES IN %.

Certificate	Argoverse 2 test			nuScenes		
	Miss	Cond.	Unnec.	Miss	Cond.	Unnec.
LTT, marginal $\alpha = 0.05$	3.7	24.2	28.5	13.2	58.1	26.1
Luo-style, $\epsilon = 0.05$	0.0	0.0	84.9	5.0	19.6	77.9
Luo-style, $\epsilon = 0.32$	4.3	28.5	26.0	14.9	66.0	21.3

S2. REVERSE SHIFT (EXPLORATORY, POST HOC)

We also certified on half of the nuScenes scenarios and deployed on Argoverse 2 test, a direction pre-registered as a reverse-shift analysis. As pre-registered, with all scenarios, the procedure *refused to certify* on both halves, because of the 5.0 % floor. On the avoidable subset (post hoc), thresholds certified on nuScenes ($n \approx 715$) give an Argoverse 2 miss rate of 1.9 % and 2.0 %, both below the target, at 40.6 % and

39.5 % unnecessary stops. A certificate from the harder dataset therefore transfers safely but conservatively, while the reverse does not transfer. This asymmetry is a single-pair observation pending replication.

S3. RECALIBRATION WITH A FEW LABELED TARGET SCENES

Table S2 draws k labeled target scenes at random and evaluates on the rest. The nuScenes analysis (avoidable subset, 60 draws) was exploratory and not pre-registered. Its Waymo counterpart (all scenarios, 200 draws) was pre-registered as R1–R4 before any Waymo outcome was read, and its outcomes are reported in Section V of the main paper. On nuScenes, uncertified tuning on 40–60 target scenes exceeds the target in about two draws in three. A certified threshold on the same scenes keeps the target (violations 0–10 %) but stops in 53–81 % of scenarios, against 27 % without target labels. A handful of labeled scenes therefore restores a *valid* certificate, not the original operating point. On Waymo, uncertified tuning violates the target in 69.5 % and 59 % of draws with 40 and 60 target scenes, and weighting the Argoverse 2 calibration set with 60 Waymo scenes ($n_{\text{eff}} = 232$) keeps validity at 0 % violation.

TABLE S2

RECALIBRATION WITH $k = 60$ LABELED TARGET SCENES. NUSCENES: EXPLORATORY, AVOIDABLE SUBSET, 60 DRAWS. WAYMO: PRE-REGISTERED (R1–R4), ALL SCENARIOS, 200 DRAWS. VIOL.: FRACTION OF DRAWS WITH TEST MISS $> \alpha$. RATES IN %.

Method	nuScenes			Waymo		
	Miss	Viol.	Unnec.	Miss	Viol.	Unnec.
AV2-certified, no labels	9.2	–	27.1	2.7	0	45.0
Target-only tuned	6.0	70	36.7	6.2	59	29.6
Target-only LTT	0.8	0	72.5	1.5	2.5	58.6
AV2 + 60, weighted LTT	2.8	0	53.3	1.7	0	49.8

S4. SIMULATOR SENSITIVITY

Three changes to the simulator, each paired by scenario seed against the main run (reactive traffic, 4.0 m/s², 2 Hz), leave every conclusion of the main paper unchanged. Non-reactive log-replay traffic raises the harm base rate (17.7 % versus the main run’s 15.1 %, since logged agents cannot react to an ego that deviates from its log) and, with it, unnecessary stops by 7.9 points [95 % CI 5.3, 10.8]. The miss rate itself does not move (paired difference -0.5 points $[-1.3, 0.0]$). A

2.0 m/s^2 comfort deceleration changes neither miss nor stops (paired difference exactly 0 in both), because both alert-level quantities depend only on the reference run and the firing tick (Observation 1 of the main paper), not on how hard the vehicle then brakes. It does lower induced collisions (2.6 % to 1.6 %), the one quantity that does depend on post-fire kinematics. Decisions at 10 Hz on 100 scenarios give a miss rate of 2.9 % [1.0, 8.2], consistent with the main run within an interval widened by the smaller sample.